

Comdyna GP-6 — Projectile Demonstration

An analog computer that flies a cannonball with wires, op-amps, and capacitors — paired with an HP 1200B oscilloscope.

Demo script · Schematic→Equation→Wiring explainer · Exhibit placard · Operator setup card · Visitor FAQ

The throughline of the whole exhibit is one idea: **the picture of the circuit *is* the math, and the wired-up machine *is* the solution being drawn in front of you.** This document contains everything needed to present, explain, run, and label the exhibit.

CONTENTS

1. [Overview & the 20-second version](#)
2. [From Schematic → Equation → Wiring \(the core explainer\)](#)
3. [Demo Script \(timed, for presenters\)](#)
4. [Exhibit Placard \(print & mount\)](#)
5. [Operator Setup Card \(run it / fix it\)](#)
6. [Visitor FAQ](#)
7. [Hardware & Sources](#)

1 Overview & the 20-second version

This machine has **no CPU and runs no software**. It solves the projectile equation $a = -g$ by *being* the equation:

- **Gravity** is a steady voltage (set by a knob).
- An **integrator** adds that up over time to get **velocity** (a straight, falling line).
- A **second integrator** adds velocity up over time to get **height** (a parabola — the flight arc).
- The **oscilloscope** plots height vs. time, so you literally watch the projectile fly.

Turn the GRAVITY knob and the arc changes *instantly* — because the wires are doing the calculus continuously, not step-by-step.

That is the charm of analog computing — math you can feel with your fingertips.

2 From Schematic → Equation → Wiring

This is the core teaching piece. It walks the same path an engineer walked in 1965 to make an analog computer solve a problem:

1. Write the physics as an equation → 2. Re-draw the equation as a block diagram (the schematic) → 3. Translate each block into a patch-cord connection on the machine.

Part 1 — The Physics

Throw a ball straight up; ignore air resistance. One force acts on it: **gravity**, a constant downward acceleration **g** ($\approx 9.8 \text{ m/s}^2$). Three quantities describe the flight, linked like a chain:

Quantity	Symbol	In plain words
Acceleration	a	How fast velocity changes. Constant here: $a = -g$ (down).
Velocity	v	Speed & direction. Starts at launch speed, drops steadily.
Position (height)	y	How high the ball is right now.

The whole problem is **one second-order differential equation**:

$$\frac{d^2y}{dt^2} = -g \quad \text{with: } \begin{array}{ll} y(0) = y_0 & \text{(launch height)} \\ y'(0) = v_0 & \text{(launch velocity)} \end{array}$$

Solved on paper it gives the familiar formulas — but **the analog computer never uses them**. It works straight from $d^2y/dt^2 = -g$ and builds the answer by integrating, the way nature does:

$$v(t) = v_0 - g \cdot t \quad \rightarrow \quad \text{a straight line sloping down}$$

$$y(t) = y_0 + v_0 \cdot t - \frac{1}{2} \cdot g \cdot t^2 \quad \rightarrow \quad \text{a parabola (the flight arc)}$$

Part 2 — The Big Trick: integrate twice

Read the chain backward — the engineer's key insight:

```

START:  acceleration = -g          (a constant – just a steady voltage)
        | integrate (add up over time)
        ▼
        velocity     = v0 - g·t    (a falling straight line)
        | integrate (add up over time)
        ▼
        position      = y0 + v0·t - ½g·t²  (a parabola – the flight)

```

So if we have a block that can **integrate**, we: (1) feed a constant “-g” in → out comes **velocity**; (2) feed velocity in → out comes **position**; (3) display position → watch it fly. That block is an **op-amp integrator**, and the GP-6 has four.

Part 3 — The Building Blocks

Each block performs **one math operation on voltages**. A voltage stands for a number (± 10 V = $\pm$ full scale; the red readout shows “percent of 10 V”, so $-0.588 = -5.88$ V).

BLOCK A — COEFFICIENT POTENTIOMETER (A KNOB = MULTIPLY BY A CONSTANT)

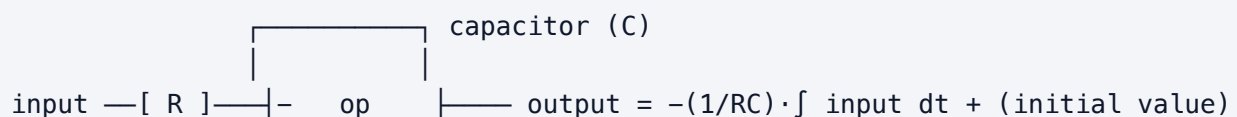
A voltage divider: feed it +10 V, get back any fraction 0–1 — that fraction *is* a number in your equation, like **g**. Here: **Pot #2 = GRAVITY**, **Pot #5 = INITIAL VELOCITY**. Turning them changes the equation’s numbers while it runs.

BLOCK B — INVERTER (MULTIPLY BY -1)

An op-amp with equal resistors flips a voltage’s sign: **out = -in**. Sign-juggling is half the job of patching. Here: **Amps #7 and #8**.

BLOCK C — INTEGRATOR (THE STAR)

An op-amp with a **capacitor** in feedback performs calculus — its output is the **running total of its input over time**:



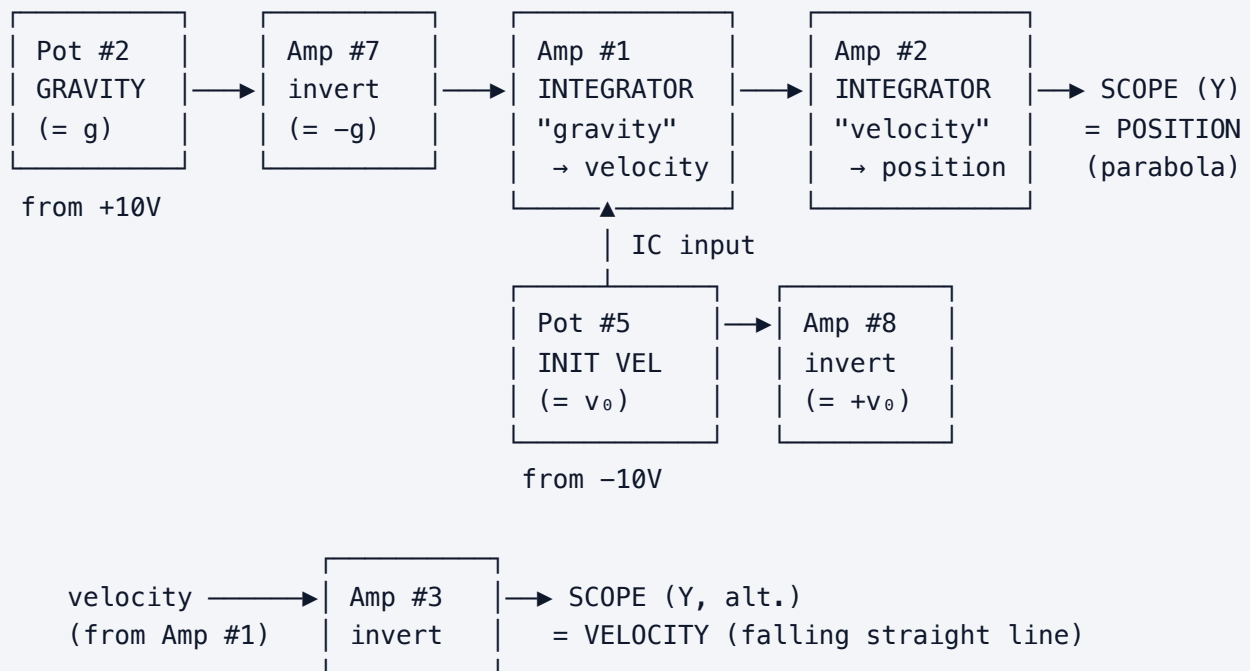


- **It accumulates.** A steady input makes the output ramp — and that ramp *is* an integral.
- **It starts from an Initial Condition (IC).** You preload the capacitor to a starting voltage. *This is how launch velocity and launch height enter the problem.*
- **It inverts as it integrates** (note the minus sign). Every integrator flips the sign of what it computes — which is why the schematic has extra inverters.

⚠ **The single most important idea for the wiring:** an op-amp integrator computes the *negative* integral, so a chain of two integrators flips the sign twice. The “extra” amplifiers (inverters #7/#8/#3) are **not** doing physics — they’re **bookkeeping for the minus signs** so velocity falls and the arc points the right way. See that, and the patch panel stops looking like spaghetti.

Part 4 — The Schematic: the equation as a wiring plan

Redraw $d^2y/dt^2 = -g$ as those blocks (this is the museum’s tan handout). Read left to right, following the signal:



Pot #2 (GRAVITY) from +10 V	the constant g
Amp #7 (inverter)	makes it -g (gravity pulls down)
Amp #1 (integrator)	first integral: $-g \rightarrow$ velocity , $v = v_0 - g \cdot t$
Pot #5 $\rightarrow$ Amp #1 IC input	the starting condition v₀ (launch speed)
Amp #8 (inverter)	fixes the sign of v_0 so launch is upward
Amp #2 (integrator)	second integral: velocity $\rightarrow$ position y
Amp #2 output $\rightarrow$ scope	the flight arc (parabola), height vs. time
Amp #3 (inverter)	display helper: flips velocity to read as a falling line

Every term in the equation has become a block; every arrow is a wire. **The schematic is the equation.**

Part 5 — The Wiring: diagram $\rightarrow$ patch cords

On the GP-6, every block's connections are brought out to color-coded jacks. "Programming" means plugging cords to make the diagram real. Each arrow above becomes one patch cord:

1. **Reference $\rightarrow$ Gravity pot.** Patch **+10 V** to **Pot #2**. Dialing it sets **g**.
2. **Gravity pot $\rightarrow$ inverter (#7).** Now you have a steady **-g** — a downward pull.
3. **-g $\rightarrow$ Integrator #1.** Amp #1 accumulates $-g \rightarrow$ output ramps. **That ramp is velocity.**
4. **Set launch speed (IC).** Patch **-10 V $\rightarrow$ Pot #5 $\rightarrow$ Amp #8 $\rightarrow$ Amp #1 IC jack.** Preloads integrator #1 so at $t=0$ its output already equals **v₀**.
5. **Velocity $\rightarrow$ Integrator #2.** Amp #2 accumulates velocity $\rightarrow$ output is **position (height)**.
6. **Position $\rightarrow$ Scope Y.** Scope X = time, so it draws **height vs. time — the parabola.**
7. **Velocity $\rightarrow$ inverter #3 $\rightarrow$ scope (optional).** Flip the display to show velocity as a falling line; *zero velocity = apogee*.

Why color-coded cords? Red carry signals; black are grounds/references; each amplifier's inputs, summing junction (SJ), and output are grouped together. Once patched, the "program" is visible — you can trace the equation with your finger.

Part 6 — Running it: IC, OP, and why the curve is alive

A digital computer would step through tiny time slices. The GP-6 doesn't step — its capacitors integrate **continuously**, at the speed of electricity. The whole flight is solved all at once, all the time. The MODE controls run the show:

Mode	What happens	Projectile meaning
PS (Pot Set)	Reference applied so you can dial & read pots on the DVM.	Set g and v_0 .
IC (Reset)	Integrators forced to starting values.	Load the launch — ball in hand.
OP (Operate)	Integrators integrate; the solution runs.	Launch! The ball flies.
HD (Hold)	Integration freezes.	Pause the ball mid-air.
RO (Repetitive Op)	Auto-cycles IC→OP many times/sec (rate = COMPUTE TIME).	Re-launches continuously → stable live trace.

Because of **Repetitive Operation**, the machine re-flies the projectile dozens of times a second and redraws the arc. So turning the **GRAVITY** knob reshapes the whole parabola in real time — flatter on the “Moon,” steeper with more gravity. Turn **INITIAL VELOCITY** and the arc grows or shrinks. Nothing is recomputed in steps; the wires simply *are* the new equation the instant you turn the knob.

THIS is the charm of analog computing.

ONE-PARAGRAPH SUMMARY (FOR MEMORIZING)

A projectile obeys $d^2y/dt^2 = -g$. Read backward, it says: integrate a constant (gravity) once to get velocity, integrate again to get height. The schematic redraws that as blocks — a knob for gravity, an integrator for velocity, a second for position, plus inverters to keep the minus signs honest. The wiring makes each block real with patch cords: reference → gravity pot → inverter → integrator #1 (velocity) → integrator #2 (position) → oscilloscope. Set launch speed by preloading integrator #1's initial condition. Run it in repetitive mode and the scope draws the flight arc live — change gravity or launch speed by hand and watch the trajectory respond instantly, because the circuit isn't *calculating* the motion, it *is* the motion.

3 Demo Script

Presenter needs: GP-6 powered on and patched (see §5), HP 1200B on and showing the trace, machine in **Repetitive Operation (RO)**.

🕒 90-Second Short Version (walk-up)

[Point at the scope's parabola.] "See that curve? That's a ball flying through the air — being calculated **right now** by this machine. But here's the thing: there's **no computer chip in here, and no software**. No ones and zeros.

This is an **analog** computer. Instead of counting, it uses **electricity to act out the physics**. Gravity is just a steady voltage — set by *this* knob. The machine adds that up over time to get the ball's speed, adds *that* up to get its height, and paints the height on the scope. That's the flight.

Watch what happens when I change gravity. *[Slowly turn the GRAVITY knob.]* **The whole trajectory reshapes — instantly**. Less gravity, the ball sails — that's the Moon. More gravity, it slams down. I didn't re-run any program. The wires just *are* the new equation the moment I turn the knob.

That's the charm of analog computing — math you can feel with your fingertips."

1. The Hook (30 sec)

[Stand beside the scope; wait for the parabola.] "Take a look at this green curve. It's an arc — like a ball thrown across a field, going up, peaking, coming back down. **And this machine is drawing it because it's computing the ball's flight in real time**. Here's what surprises people: **this is a computer with no chip, no code, and no software**. It's called an **analog computer**."

2. Digital vs. Analog — the core idea (60 sec)

"Your phone is a **digital** computer — it solves problems by **counting**, billions of yes/no steps a second. To draw this arc it would chop the flight into thousands of slices. This machine is **analog**. It doesn't count — it uses **voltages to physically stand in for real quantities**. A voltage here *is* gravity, another *is* the ball's speed, another *is* its height. Instead of running a program, it's **wired** so the electricity obeys the same physics the real ball obeys. **It doesn't calculate the motion — it becomes the motion.**" *[Gesture to the*

patch cords.] "Those wires are the 'program.' Re-plug them and you've changed the equation."

3. The physics, in three steps (75 sec)

[Mime a ball arcing.] "Only one thing acts on a thrown ball: **gravity**, pulling down steadily. So: **One** — gravity is a steady voltage, set by this knob marked **GRAVITY**. *[Tap Pot #2.]* **Two** — add up that pull over time and you get the ball's **speed**; the block that adds up over time is an **integrator**, and this machine has four. *[Point to amps 1–4.]* **Three** — add up the *speed* over time and you get the ball's **height**; that takes a *second* integrator, and its output is the arc on the screen. So the recipe is: **gravity → integrate → speed → integrate → height → draw it.**"

4. Read the wiring like a sentence (60 sec)

[Point to the tan schematic.] "This diagram is the equation drawn as a wiring plan, and the panel matches it. Follow my finger: start at the **GRAVITY knob** → into the **first integrator** → out comes **speed**. This other knob, **INITIAL VELOCITY**, sets how hard we throw it. *[Tap Pot #5.]* Into the **second integrator** → out comes **height** → to the **scope**, plotted against time. That's your arc." *[Optional]* "Those extra amps labeled 'invert' don't do physics — they just **flip plus and minus signs** so gravity pulls down and the throw goes up."

5. The Payoff — make it live (90 sec)

"Here's the magic. This machine is **re-flying the ball dozens of times a second**, so I can change the world **while the ball is in flight.**" *[Slowly turn GRAVITY down.]* "Watch as I **weaken gravity**... the arc **stretches** — the ball sails and hangs longer. **That's a throw on the Moon.**" *[Turn GRAVITY up past normal.]* "Now **crank gravity up** — the arc **collapses**. Heavy planet." *[Turn INITIAL VELOCITY up.]* "Throw it harder — bigger arc." *[Pause.]* "I never typed anything or re-ran a program. **The trajectory changes the instant I turn the knob — because the wiring is the equation.**" *[Optional, switch to velocity view.]* "This straight line falling through zero is the ball's **speed**. Where it crosses zero, the ball has stopped rising — **that instant is the apogee, the very top**. After that, it falls."

6. The Close (30 sec)

"For about thirty years — through the Apollo program, aircraft and missile design — **this** is how engineers solved the hardest equations: not by counting, but by building a little electrical model and watching it run. Digital eventually won on precision and flexibility. But

for *feeling* how a system behaves, nothing beats it. **That's the charm of analog computing.**"

Presenter notes & recovery lines

- **Flat / off-screen trace:** likely in IC/HOLD or a pot at zero — confirm RO/Operate and that GRAVITY & INIT VEL aren't fully down (see §5).
- **"Is it broken? Where's the screen/keyboard?"** — "There isn't one. You program it with **wires**, and the answer comes out as a **picture** on the scope. The wires are the keyboard."
- **"Why does the readout say -0.588?"** — "A voltmeter you can point at any wire; it reads in *percent of full scale*, so $-0.588 \approx -59\%$."
- **"Could I build one?"** — "Yes — the heart is an op-amp and a capacitor, a few dollars of parts."
- Keep hands **slow** on the knobs — the magic is watching the curve morph. Don't re-patch cords mid-tour unless confident.

The one line to never skip: "It doesn't calculate the motion — it *becomes* the motion. That's the charm of analog computing."

4 Exhibit Placard (print & mount)

Watch a Cannonball Fly — Computed by Wires, Not Code

Comdyna GP-6 Analog Computer · with HP 1200B Oscilloscope

The green arc on the screen is a **projectile in flight** — and this machine is computing it **right now**. But look closer: there is **no microchip and no software** inside. This is an **analog computer**. Instead of counting in ones and zeros, it uses **voltage** to stand in for real things. One voltage *is* gravity. Another *is* the projectile's speed. Another *is* its height. The machine is **wired** so the electricity obeys the very same physics a real cannonball obeys.

It solves the motion by **integrating** — continuously adding gravity up into speed, and speed up into height — using op-amps and capacitors. The answer is drawn live on the oscilloscope: **height versus time, the flight arc**.

HOW IT THINKS

GRAVITY (a knob = a voltage)
| add up over time (integrate)
▼
SPEED (a falling straight line)
| add up over time (integrate)
▼
HEIGHT (a parabola – the flight on the screen)

The wiring *is* the equation. Re-plug the patch cords and you've changed the physics.

TRY THIS (DOCENT-ASSISTED)

Turn the GRAVITY knob and watch the arc change shape — instantly. Weaker gravity → the projectile sails (the Moon). Stronger gravity → it slams down. Nothing is re-run; the trajectory simply *becomes* the new equation the moment you turn the knob.

That is the charm of analog computing — math you can feel with your fingertips.

DID YOU KNOW?

For about **30 years** — through the Apollo program, aircraft and missile design — analog computers like this solved the hardest equations: not by counting, but by building a small **electrical model** of the problem and watching it run, at the speed of electricity.

Analog computers represent numbers as voltages (here, ± 10 V = full scale) and solve differential equations with op-amp integrators. The Comdyna GP-6 was a widely used teaching machine of the 1970s–80s, programmed entirely by patch cords — the visible tangle of wires is the “source code.”

5 Operator Setup Card

Authority: the museum's **tan schematic handout** is the master wiring diagram for this specific patch. If anything here disagrees with the physical panel or that sheet,

trust the sheet. The jack list below reconstructs that schematic block-for-block — verify against the panel before re-patching.

A. Power-up & display

1. Power on the GP-6 and the **HP 1200B**; warm up ~1 min.
2. Scope: computed signal on **vertical (Y)**, **time on horizontal (X)**. Start **~1 V/div**, sweep so one flight fills the screen; adjust intensity/focus.
3. Confirm the red **OVLD** lamp is **off**. If lit, an amp is railed — back off the GRAVITY/INIT-VEL pots and recheck patching.

B. Front-panel settings (the photographed configuration)

Control	Setting	Why
MODE SELECTOR knob	OPR	Lets run modes / repetitive operation drive the solution.
Run mode	RO (Repetitive)	Auto-cycles IC→OP → live, stable trace.
COMPUTE TIME knob	mid (~ 20–50)	Repeat rate / run length. Tune so the arc is steady and fills the screen.
GRAVITY — Pot #2	~ 0.16 (≈ -0.160 DVM)	The value of g . The “hero” knob.
INITIAL VELOCITY — Pot #5	~ 0.66 (≈ -0.660 DVM)	Launch speed v_0 . Second demo knob.

DVM ADDRESSING (AS LABELED ON THE PANEL TAPE) — READS PERCENT OF 10 V

Selector	Setting	Reads / means
X ADDRESS	TIME	Horizontal = time base (drives sweep).
Y / POT ADDRESS	7 = GRAVITY, 5 = $G \times V$	Meter at the gravity term / gravity $\times$ velocity node.
Solution address	3 = VELOCITY, 2 = POSITION	Amp #3 = velocity line; Amp #2 = position parabola.

To switch the scope between parabola (position, Amp #2) and falling line (velocity, via Amp #3), move the Y display selector to that amplifier — the schematic calls this “Select 4 on the Y switch” to view the descending velocity line. Confirm your scope’s Y-address once and mark it.

C. Patch list (block-for-block)

Each line = one signal hop. Jack groups: each amp has **SJ** (input), **IC** (initial-condition input), **output**; pots have input + wiper/output; references are **+REF / –REF (± 10 V)**.

#	From	To	Represents
1	+10 V REF	Pot #2 input (GRAVITY)	constant to scale into g
2	Pot #2 output	Amp #7 input (inverter)	makes $-g$
3	Amp #7 output	Amp #1 SJ (integrator)	$-g \rightarrow$ 1st integrator $\rightarrow$ velocity
4	–10 V REF	Pot #5 input (INIT VEL)	constant to scale into v_0
5	Pot #5 output	Amp #8 input (inverter)	makes $+v_0$ (upward)
6	Amp #8 output	Amp #1 IC input	preload integrator #1 $\rightarrow$ start vel = v_0
7	Amp #1 output	Amp #2 SJ (integrator)	velocity $\rightarrow$ 2nd integrator $\rightarrow$ position
8	Amp #2 output	Scope vertical (Y)	height vs time = the parabola
9	Amp #1 output	Amp #3 input (inverter)	sign-flip velocity for display
10	Amp #3 output	Scope vertical (Y, alt.)	velocity = falling straight line
—	Time base / ramp	Scope horizontal (X)	sweep = time axis

Grounds: black cords tie scope and computer references together as shown on the sheet — don’t remove them thinking they’re spares.

D. Run sequence (each demo)

1. MODE SELECTOR $\rightarrow$ **POT SET**; confirm GRAVITY (#2) and INIT VEL (#5) on the DVM; return $\rightarrow$ **OPR**.
2. Ensure run mode is **RO**; the scope shows a steady arc.
3. Demo the **GRAVITY** knob (slow turns): arc stretches (Moon) / collapses (heavy planet).
4. Demo the **INITIAL VELOCITY** knob: bigger/smaller arc.

5. To freeze for discussion: tap **HD**; resume with **OP/RO**.

E. Troubleshooting

Symptom	Likely cause	Fix
OVLD lit, trace pinned to edge	pot too high / amp saturated / mis-patch	Lower GRAVITY & INIT-VEL; verify cords 1–7 vs. sheet.
Flat line, no arc	in IC/HOLD, pot at zero, or scope on wrong node	Set RO/OPR; raise pots; confirm scope Y on Amp #2.
Arc drifts / won't sit still	COMPUTE TIME or scope sweep mismatched	Re-tune COMPUTE TIME & time/div until one flight fills screen.
Curve frozen, ignores knobs	stuck in HOLD, or MODE in POT SET	Return to OPR + RO.
Wrong trace (line vs. parabola)	scope Y on wrong amp	Amp #2 = parabola; Amp #3 = velocity line.
Readout ~0 for a live signal	DVM addressed to idle node	Re-address Y/POT or Solution selector (2=position, 3=velocity).

F. Numbers cheat-sheet

- **Full scale:** $\pm 10\text{ V} = \pm 100\%$. DVM shows percent of 10 V ($-0.160 \rightarrow -16\% \rightarrow -1.60\text{ V}$).
- **Integrator law:** $\text{output} = -(1/RC) \times (\text{running total of input}) + \text{initial condition}$. The minus sign is why inverters (#7, #8, #3) exist.
- **Apogee:** where the velocity line crosses zero = the top of the flight.

6 Visitor FAQ

Q: Where's the screen and keyboard? Is it broken?

Nothing's broken — there isn't one. You program it with **patch cords**, and the answer is a **picture on the oscilloscope**. The wires are the keyboard; the scope is the screen.

Q: Is there a computer chip in here?

No. No chip, no memory, no software, no ones and zeros. The brains are **op-amps and capacitors** wired to act out the physics directly. It's an *analog* computer.

Q: Analog vs. digital, really?

Digital **counts** — billions of tiny yes/no steps. Analog **measures** — a smoothly varying voltage stands for a real quantity. Here a voltage *is* gravity, *is* speed, *is* height. Digital approximates by counting fast; analog mirrors the real thing continuously.

Q: What is it computing?

The flight of a projectile under gravity. The green arc is its **height over time**; switch the view to see its **speed**.

Q: How does it do math without code?

By **integrating** — continuously adding up. Gravity (a steady voltage) adds up over time into **speed**; speed adds up into **height**. Two integrations, done by op-amp-and-capacitor **integrators**.

Q: Why does turning the knob change the curve instantly?

The **wiring itself is the equation**, and it re-flies the projectile dozens of times a second. Turn a knob and you've changed a number in the live equation, so the arc reshapes immediately and smoothly.

Q: What do the two knobs do?

GRAVITY sets how strong gravity is (down = arc stretches like the Moon; up = drops fast). **INITIAL VELOCITY** sets how hard you throw it.

Q: What's the red number?

A built-in voltmeter you aim at any wire. It reads in **percent of full scale** (full scale = 10 V), so $-0.588 \approx -59\%$.

Q: What are the triangles labeled "invert"?

Sign-flippers. Each integrator flips plus/minus as a side effect, so little **inverter** amps put the signs back — gravity pulls down, the throw goes up.

Q: Where's the top of the flight?

Switch to the **velocity** view — the straight line falling through zero. **The zero-crossing is the apogee**.

Q: Real work, or a teaching toy?

Both. This model (GP-6) was a teaching machine, but big analog computers were serious tools for ~30 years — **Apollo, aircraft and spacecraft design, missile guidance, control systems** — because they solve differential equations instantly and let you feel a system respond.

Q: Why did digital win?

Precision, repeatability, memory, flexibility. But for **intuition** — watching a whole system

breathe as you turn a knob — analog is still wonderful, which is why people build them today.

Q: Could I build one at home?

The core is just an **op-amp and a capacitor** — a few dollars of parts. This box simply packages eight amplifiers, multipliers, and a tidy patch panel.

Q: How accurate is it?

Roughly a percent — enough to see the physics, not to do banking. Trading some precision for **speed and insight** was the whole point.

Leave them with: “It doesn’t *calculate* the motion — it *becomes* the motion. That’s the charm of analog computing.”

7 Hardware & Sources

Comdyna GP-6 general-purpose analog computer (≈1970s–80s teaching machine): 4 integrator/summer amplifiers (top, labeled 1–4), 2 inverter amplifiers (7–8), 2 multipliers, 8 coefficient potentiometers, ± 10 V & ± 15 V references, built-in addressable digital voltmeter (the red readout), capacitor-shortening for zeroing integrators, overload protection. Paired with an **HP 1200B oscilloscope** (X = time, Y = computed quantity). The “program” is the patch cords.

- [Comdyna GP-6 — Carleton University, Vintage Computing](#)
- [GP-6 Analog Computer Operator’s Manual \(VTDA, PDF\)](#)
- [Comdyna GP-6 Operator’s Manual \(bitsavers, PDF\)](#)
- [Comdyna GP-6 — Old Computer Museum](#)
- [JonDent — Comdyna GP-6: introduction to the patch bay](#)
- [WA6PZB — Projectile Motion with Analog Computers](#)
- [All About Circuits — Op-Amp Integrator lab](#)

Comdyna GP-6 Projectile Demonstration — Exhibit Kit. Prepared for the museum exhibit pairing the GP-6 analog computer with the HP 1200B oscilloscope. The patch list reconstructs the museum’s tan schematic

block-for-block; verify against the physical panel and that sheet before re-patching.